

Mean-Reversion Portfolio Selection on US-Listed ETFs, 1999–2026: A Replication with Pre-Registered Tests

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Abstract

Lahanis et al. (2025) report that online portfolio selection algorithms built on mean reversion, CWMR and PAMR among them, multiply wealth by hundreds to over a thousand times on NASDAQ-100 stocks over 1998–2009. We reproduce their tables exactly from their published code and data, then test the strongest algorithm, the authors’ variant of Confidence-Weighted Mean Reversion (CWMR), on 16 liquid US-listed equity ETFs with measured spreads and fills that could have been executed. Daily CWMR trades about 350 times its account a year and trails SPY at measured 2016–2024 spreads. We searched variants that trade less, selected one on 2016–2024 data and tested it on data it had not seen. A monthly variant selected this way matched SPY on a pre-registered 2025–2026 holdout (18.28% against 18.36% a year). Variants that learn from weekly or monthly bars passed a pre-registered 1999–2015 test against SPY: weekly bars returned 9.82% a year against SPY’s 4.90%. That test turned out to be weak. An equal-weight portfolio of the same ETFs returned 6.61%, and the strategy’s Sharpe ratio is not significantly higher than it (block-bootstrap $p = 0.22$). Its lead over equal weight disappears at about 4.5 basis points per dollar traded, and most of it disappears if each trade is made one session later. We have no measured spreads for these ETFs before 2016. We find no statistically significant edge over simply holding the same ETFs.

1 Introduction

Online portfolio selection treats investing as a repeated game. Each period an algorithm holds a portfolio, observes the period’s price changes and updates its weights by a fixed rule. Cover (1991) showed that one such rule, the universal portfolio, achieves asymptotically the same growth rate as the best constant-rebalanced portfolio chosen in hindsight. A later family of algorithms bets on reversal instead. Anticor (Borodin et al., 2004), PAMR (Li et al., 2012), CWMR (Li et al., 2011, 2013) and RMR (Huang et al., 2013) move weight from recent winners to recent losers. The papers that introduced them report very large wealth multiples on a common set of benchmark datasets; Li and Hoi (2014) survey the family.

Lahanis et al. (2025) implement about 25 of these algorithms and test them on NASDAQ-100 stocks from 1998 to 2009. Their best result, CWMR, multiplies wealth 1,590 times. They disclose that they ignore trading costs and that the period may not represent recent markets.

The question here is whether any version of their CWMR survives trading costs and timing on instruments a retail investor can buy today. The backtests use `btest`, the author’s backtesting software, which times orders the same way as the author’s paper-trading system.

Related work. Trading costs have been studied in this literature from the start. Blum and Kalai (1999) extend universal portfolios to proportional costs, Li et al. (2011) report that CWMR withstands small costs on their benchmark datasets, Li et al. (2018) design an algorithm that trades reversion against costs, and Li et al. (2016) publish a toolbox for these experiments. Closest to this paper, Moon et al. (2019) test PAMR, OLMAR and cost-optimised variants on S&P 500 stocks over 2000–2017; they find that the standard benchmark datasets favour mean reversion and that the strategies may fail when explicit or implicit costs exist. At the level of assets, short-horizon reversal is documented for individual stocks at weekly (Lehmann, 1990) and monthly (Jegadeesh, 1990) horizons, while industry portfolios show momentum (Moskowitz and Grinblatt, 1999), so the stock-level evidence does not carry over directly to sector ETFs. Part of measured short-term reversal is bid-ask bounce (Roll, 1984), and the returns to reversal strategies can be read as pay for supplying liquidity (Nagel, 2012).

This paper adds three things: an exact reproduction of a recent paper that found implementation defects in it; a test of its best algorithm on instruments a retail investor can trade, with measured spreads and fills that could have been executed; and confirmatory tests specified before they were run, with the deviations from that plan reported. Section 2 describes the data and methods. Section 3 reproduces the original paper. Sections 4 to 6 report the out-of-sample tests in the order we ran them, since the order matters for how much each result can be trusted. Section 7 reports where we departed from the pre-registered plans. Section 8 discusses what the evidence supports.

2 Data and methods

2.1 Algorithm

All tests use the authors’ CWMR as written in their public repository, ported to btest and checked against their function to within 7.1×10^{-9} per weight on their data. Their code describes it as a simplified version of the published algorithm (Li et al., 2011, 2013). Among other differences, its step size has no confidence term, a learning-rate factor η is added and the update is not mean-centred. We keep their parameters throughout ($\epsilon = 0.89$, $\theta = 0.92$, $\eta = 0.93$) and never re-tune them.

Each period the algorithm receives a vector x of price ratios, one per asset ($x_i = 1.02$ means asset i rose 2%). With current weights b and covariance Σ (initially the identity), it computes

$$\lambda = \eta \max\left(0, \frac{b^\top x - \epsilon}{x^\top \Sigma x}\right), \quad (1)$$

$$\mu = b - \lambda \Sigma x, \quad (2)$$

$$\Sigma^{-1} \leftarrow \Sigma^{-1} + 2\lambda\theta xx^\top, \quad (3)$$

$$b \leftarrow \Pi_\Delta(\mu), \quad (4)$$

where Π_Δ projects onto non-negative weights that sum to one. Because $\epsilon = 0.89$, the update fires unless the portfolio falls more than 11% in one period, so in practice it fires every period. Assets that rose most lose the most weight.

We test four schedules (Table 1).

Table 1: CWMR schedules tested.

Name	Learns from	Trades
Daily	daily ratios	every session
Monthly, daily signal	daily ratios	last session of each month
Monthly bars	month-end to month-end ratios	last session of each month
Weekly bars	week-end to week-end ratios	last session of each ISO week

2.2 Universe and data

The universe is 16 ETFs: SPY, QQQ, IWM, DIA, EFA, EEM and ten Select Sector SPDRs (XLK, XLF, XLY, XLE, XLI, XLP, XLU, XLB, XLRE). EFA and EEM hold non-US stocks and XLRE holds real estate. The ETFs launched between 1993 and 2015, so the universe grows over time: it has 11 members in January 1999 and reaches 16 in October 2015. A fund joins on its first trading day. Before that it holds no weight. Until a fund trades, the algorithm gives it the average price ratio of the funds that do and splits its weight equally among them, which gives the same portfolio return, so the algorithm runs on the whole universe from the start and keeps what it has learned when a fund joins. The universe is a set of large, liquid ETFs chosen in 2026. Funds that existed in 1999–2015 and later closed or shrank are not in it, which favours any strategy that holds these funds, including the equal-weight benchmark.

We use two price sources. From 2016 on, btest uses Alpaca SIP minute bars adjusted for splits and dividends; a strategy decides at 15:30 New York time on data through 15:14 and fills at 15:45, which is how the author’s paper-trading system trades. For runs that start before 2016 we use Yahoo daily bars adjusted for dividends and splits; a strategy decides on a session’s close and fills at the next session’s open. The stored Yahoo closes match a fresh download to within a ratio of 3×10^{-6} on every date.

2.3 Costs and benchmarks

Every dollar traded pays a flat cost in basis points (bp), plus the SEC fee on sales (\$20.60 per million dollars sold). Turnover is dollars bought plus dollars sold per year, divided by average equity. We measured half-spreads at 15:45 from Alpaca quotes on 45 sessions in 2016–2024 (Table 2).

Table 2: Median half-spreads at 15:45, 2016–2024.

ETF	Half-spread (bp)
SPY	0.17
QQQ, IWM, DIA	0.26–0.30
Most sector SPDRs, EFA	0.41–0.82
EEM, XLRE, XLF	1.20–1.68
Equal-weight average	0.70

We charge 0.7 bp unless stated and report results at 5, 10 and 20 bp. The model has no commissions, no market impact and no taxes; Section 8 discusses each.

The benchmarks are SPY with dividends reinvested and an equal-weight portfolio (EW) of the same ETFs, reset at each month end. EW differs from the constant rebalanced portfolio (CRP) of Section 3, which resets every day. EW matters more. A strategy that beats SPY only because these ETFs beat SPY has learned nothing.

2.4 Statistics and procedure

We compare Sharpe ratios with the test of Jobson and Korkie (1981) with the correction of Memmel (2003), on daily returns of two strategies over the same days. The test uses their correlation, 0.71 to 0.98 across this study, so it can detect smaller differences than comparing two independent estimates. It assumes independent, normally distributed returns, and these daily returns have fat tails and clustered volatility. We therefore also report a moving-block bootstrap (Künsch, 1989) with 21-day blocks and 10,000 draws, with a 95% interval for the annual Sharpe difference; Ledoit and Wolf (2008) discuss related robust tests. The tests compare Sharpe ratios of raw returns; the tables report Sharpe ratios in excess of the 3-month Treasury bill rate.

Each confirmatory test was written into the research log and committed to git before it ran. The commit names the strategy version, window, settings and pass criterion, and states that the strategy will not be changed and rerun on that window. The texts of the three plans are in the replication repository, each with its commit hash and time, copied word for word. Git commit times come from the author’s machine. For the holdout test, GitHub’s own record shows the plan arriving 26 seconds before the run was created. For the other two plans the order rests on the repository history, which can be shown to referees on request. We kept 2025-01-01 onward as a holdout. Two unrelated single-ETF strategies had been run on it before this study; no portfolio selection strategy had, and we ran one test on it.

3 Reproducing the original paper

The authors’ repository contains their data, 93 NASDAQ-100 stocks from 1998-01-02 to 2009-12-31. Running their functions reproduces their Tables 3 and 4 exactly: CWMR 1,590.38×, PAMR 788.26×, Anticor 36.89×, FTRL 15.25× and the equal-weight constant rebalanced portfolio (CRP) 12.16×.

The paper describes walk-forward validation of its parameters, but the validation windows lie inside 1998–2009 and its Tables 3 and 4 report the whole sample, so these are in-sample results. The paper gives its sample as 1998–2010; the data in the repository ends on 2009-12-31. Within the sample, mean reversion keeps much of its return under frictions (Table 3).

Table 3: The authors’ data, 1998–2009. Paired Sharpe test against CRP.

	CAGR	p vs CRP
CRP	23.2%	
CWMR	85.0%	< 0.0001
CWMR, 10 bp	43.6%	0.29
CWMR, one-day delay	45.6%	0.11
PAMR, one-day delay	51.9%	0.0002
PAMR, one-day delay and 5 bp	33.9%	0.49

Four of the paper’s strategies differ from the algorithms they are named after. As implemented, OLMAR and exponential gradient never change their target weights. Both reduce to CRP, and the paper reports identical numbers for all three. CWMR and RMR are modified versions of the published methods. A fifth result depends on the data: FTRL’s lead over CRP disappears when the 32 tickers with incomplete data are removed. Two bad prints, PCAR in February 2000 and XRAY in July 2004, account for about 30% of CWMR’s final wealth.

4 Daily trading, 2016–2024

On the 16 ETFs, deciding at 15:30 and filling at 15:45, daily CWMR returns 16.6% a year before costs and 13.8% at 0.7 bp, against SPY’s 14.5%. It turns over about 350 times its account a year, so each basis point of cost takes roughly 3.5 percentage points off the annual return. Its break-even cost against SPY lies between 0.43 and 0.70 bp. Paired tests against SPY are not significant at any setting we tested ($p = 0.51$ for the paper’s close-to-close timing before costs). Daily CWMR is not worth pursuing at retail costs, and the remaining sections test schedules that trade less.

5 Selecting a lower-turnover variant, and the holdout

We ran the paper’s 13 strategy families on 2016–2024 with daily trading, with a daily signal and monthly trades, and (for 12 of them) with monthly bars: 38 configurations. CWMR with a daily signal and monthly trades did best, 18.3% a year (Sharpe 0.87) against SPY’s 14.5% (0.73), with turnover cut to $18\times$. Its paired p against SPY was 0.24, and it was the best of 25 lower-turnover configurations, so we treated it as a hypothesis and pre-registered a test on the 2025–2026 holdout. It matched SPY’s return (Table 4).

Table 4: Pre-registered holdout, 2025-01-01 to 2026-10-02.

	Monthly CWMR	SPY
CAGR	18.28%	18.36%
Sharpe (T-bill)	0.81	0.87
Max drawdown	−20.3%	−18.7%

The paired p was 0.77, and the strategy beat SPY in 8 of 21 months. With 438 sessions the test cannot tell a small edge from none, so the result is inconclusive rather than a tie. A later continuous run showed why the 2016–2024 result was misleading. The 18.3% depends on starting the algorithm fresh in January 2016. Run on daily ETF data from 1995 with its state carried forward, the same strategy returns 11.2% over 2016–2024 against SPY’s 14.4%.

6 Learning from weekly and monthly bars, 1999–2024

The monthly variant above learns from daily moves but holds for a month. Short-horizon reversal in stock returns is documented over weeks (Lehmann, 1990) and a month (Jegadeesh, 1990), so a signal built from one day’s moves and held for a month measures one horizon and bets on another. Variants that learn from week-end or month-end prices remove that mismatch. We proposed them after the holdout result above. On 2016–2024 minute data, with the algorithm started in January 2016, they returned 17.6% (weekly bars) and 11.6% (monthly bars) against SPY’s 14.5%. Weekly bars beat SPY there only with calendar weeks: started on each of the five possible offsets of 5-session blocks instead, none of the five did (9.4% to 13.9%). Neither variant had been run on data before 2016. The 1999–2015 window had been seen only by the monthly daily-signal variant of Section 5.

6.1 Pre-registered test, 1999–2015

We pre-registered both variants on 1999–2015 with ETF-only daily data. The pass criterion was CAGR and Sharpe ratio above SPY’s at 0.7 bp. This period contains two crashes, and SPY

returned only 4.9% a year (Table 5).

Table 5: Pre-registered test, 1999–2015, at 0.7 bp per dollar traded. CAGR in % a year; Sharpe ratio over T-bills; turnover per year.

	CAGR	Sharpe	Max drawdown	Turnover
Weekly bars	9.82%	0.45	−50.1%	79×
Monthly bars	8.90%	0.40	−61.4%	14×
Equal weight (EW)	6.61%	0.33	−54.4%	0.3×
SPY	4.90%	0.25	−55.2%	0×

Table 6: Paired Sharpe ratio tests, 1999–2015: annual Sharpe difference, Jobson–Korkie–Mennel p , block-bootstrap p and 95% bootstrap interval.

	Difference	JKM p	Bootstrap p	95% interval
Weekly bars vs SPY	+0.19	0.11	0.026	[+0.03, +0.37]
Weekly bars vs EW	+0.11	0.33	0.22	[−0.06, +0.29]
Monthly bars vs SPY	+0.14	0.17	0.070	[−0.01, +0.30]
Monthly bars vs EW	+0.06	0.52	0.42	[−0.08, +0.20]
EW vs SPY	+0.08	0.10	0.022	[+0.01, +0.16]

Both variants pass the criterion as written. But equal weight also passes it, and neither variant’s Sharpe ratio is significantly higher than equal weight’s (Table 6). Against SPY, weekly bars is significant under the bootstrap but not under the Jobson–Korkie–Mennel test, and so is equal weight itself. In hindsight the criterion was too easy. We already knew this basket of ETFs had beaten SPY over this window when we wrote it, so equal weight was the right benchmark to name.

6.2 Robustness to the day the week ends

Weekly bars end on each calendar week’s last session, usually a Friday. We reran the strategy with blocks of 5 sessions at each of the five possible starting offsets instead; holidays make these blocks drift against calendar weeks. Each version ran once over 1999–2024, with the algorithm learning from 1995 (Table 7).

Table 7: Weekly bars by where the week ends, at 0.7 bp. CAGR in % a year, each period taken from one run over 1999–2024.

	1999–2015	2016–2024	1999–2024
Calendar weeks (Friday)	9.9%	16.4%	12.1%
Five 5-session offsets	7.8–13.4%	11.6–14.8%	9.6–13.5%
Equal weight	6.6%	11.8%	8.4%
SPY	4.9%	14.4%	8.1%

Before 2016 every version beats both benchmarks. After 2016 only the calendar version and one offset beat SPY. The calendar version’s 16.4% over 2016–2024 differs from the 17.6% above because here the algorithm has learned since 1995 and trades on daily data. Over the whole window

the calendar version is the second best of six, and its 2016–2024 result is the best, which is what selection on 2016–2024 would produce.

The weekly variant is also concentrated. After rebalancing it holds a median of 2 ETFs, and one ETF carries 90% or more of the account in 36% of weeks.

6.3 Costs

At $79\times$ turnover over 1999–2015 and $76\times$ over 1999–2024, each basis point of cost removes about 0.8 percentage points a year (Table 8).

Table 8: Weekly bars at higher costs, in bp per dollar traded. CAGR in % a year.

	0.7 bp	5 bp	10 bp	20 bp	SPY
1999–2015	9.82%	6.24%	2.23%	−5.35%	4.90%
1999–2024	12.10%	8.50%	4.46%	−3.17%	8.09%

The break-even against SPY is about 6–7 bp over 1999–2015 and about 5–6 bp over 1999–2024. Against equal weight it is lower: over 1999–2015 weekly bars returns 7.06% at 4 bp and 6.65% at 4.5 bp, against equal weight’s 6.61%. Our spread measurements cover 2016–2024 at 15:45; we have none for these ETFs before 2016 or at the open, so we cannot say whether costs in 1999–2015 were below 4.5 bp. Estimating them from daily prices does not work at this scale. We applied the efficient estimator of Ardia et al. (2024), which its authors show improves on the earlier high–low estimators of Corwin and Schultz (2012) and Abdi and Ranaldo (2017), to each ETF’s daily open, high, low and close in each year. On 2016–2024, where half-spreads were measured at 0.17 to 1.68 bp, its yearly estimates range from −34 to +74 bp and its medians by ETF from −11 to +17 bp; for SPY it gives 11.9 bp against a measured 0.17 bp. Daily price noise swamps spreads this small, so the estimates for earlier years cannot be used. Over 1999–2001 the strategy also made 243 to 315 trades a year. At the 1999 average online commission of about \$16 a trade, that is a fifth to a quarter of a \$20,000 account each year.

6.4 Trading one session late

If the weekly lead came from a reversal that unfolds over days, delaying each trade by one session should cost little. Filling at the open two sessions after the signal instead of one lowers weekly bars over 1999–2015 from 9.82% to 7.36% a year (Sharpe 0.35), and the bootstrap p against equal weight rises from 0.22 to 0.92. Most of the lead comes from the first session after the signal.

7 Deviations from the pre-registered plans

Two procedures departed from the pre-registered plans. First, the registered design for the 1999–2015 test of Section 6 decided and filled at each session’s close, so each order filled at the closing price its decision had used, which no one can trade at. We moved fills to the next session’s open and reran both variants once with nothing else changed. Before the correction weekly bars returned 11.50% a year (paired $p = 0.02$ against SPY) and monthly bars 8.47%. Both passed the criterion before and after; Table 5 reports the corrected run.

Second, an earlier pre-registered test of the monthly variant on 1995–2015 filled the years before each ETF launched with similar mutual funds and indexes, and used the same uncorrected fills. It passed, returning 11.57% a year against SPY’s 9.33%. We no longer rely on it. With each ETF’s

own history only and corrected fills, the same strategy returns 6.79% over that window against SPY’s 9.33%. With SPY as its only fund it has nothing to choose between and stays in cash; its first trade, into DIA, is in January 1998.

8 Discussion

The original paper’s results are real, in the sense that its code computes them correctly from its data. They do not survive the move to instruments and costs a retail investor faces today. Every schedule we tried either trails SPY at measured costs (daily), matches it on a short holdout (monthly), or beats SPY by a margin the tests cannot separate from simply holding the same ETFs in equal weights (weekly and monthly bars, 1999–2015).

The weekly-bars variant is the most interesting result. Before 2016 it beat both benchmarks at every offset. Several explanations fit. Weekly reversal across sector ETFs may have been real in a market with wider spreads and slower arbitrage, in which case the profit belonged to whoever supplied liquidity (Nagel, 2012) and was not available to someone paying the spread. Daily returns of these ETFs in excess of SPY show negative next-day autocorrelation before 2016 (median across the ETFs -0.065 in 1999–2003, -0.092 in 2004–2008 and -0.047 in 2009–2015) and none after (0.000 in 2016–2024). That pattern fits bid-ask bounce in closing prints (Roll, 1984), which a backtest captures and a trader pays for, and so does the drop when trades are made one session late. And the variant came out of a search over about 100 configurations across this study (an approximate count), which inflates the best result (Harvey et al., 2016). We have not corrected for it; a deflated Sharpe ratio (Bailey and López de Prado, 2014) would be the next step. We cannot separate these explanations with the data we have. Each predicts the effect fades as markets become cheaper to trade, and after 2016 it roughly ties SPY.

Before calling it an edge we would need:

1. spreads for these ETFs before 2016 measured from historical quote data, to replace the flat cost;
2. a pre-registered test against equal weight, not SPY, on data no variant has touched. The 2025 holdout has been used once; data from 2027 on would be clean;
3. results after commissions for a stated account size, and after tax for taxable accounts. At $76\times$ turnover almost every gain is short-term.

9 Conclusion

We reproduced a published set of mean-reversion results exactly, then tested the strongest algorithm on liquid ETFs. The variant selected on 2016–2024 matched SPY on a short holdout. Weekly and monthly variants beat SPY over 1999–2015 in a pre-registered test, but an equal-weight portfolio of the same ETFs passes that test too, and neither variant’s Sharpe ratio is significantly higher than equal weight’s. The weekly lead over equal weight needs costs below about 4.5 bp and mostly disappears if trades are made one session later. We find no statistically significant edge for this CWMR variant over holding the same ETFs.

Data and code availability

Code that reproduces Section 3, Tables 3 and 5–8, the one-session-late result, the spread estimates and the autocorrelations from public data, together with the texts of the three pre-registered plans, is at

<https://github.com/yelgabo/cwmr-etf-replication>

It uses the original authors’ public code and data (Lahanis et al., 2025), Yahoo Finance daily prices and the 3-month Treasury bill rate from FRED. Sections 4 and 5 used minute bars from Alpaca, which require an account; that code is available from the author on request.

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